

PUA 4×4

Font-generation evidence · v0.4 release candidate 1

From approved mathematics to stored TrueType data, Linux selection, fractional raster behavior and a real terminal capture

RESULT

Candidate.3 preserves the approved mapping for all 65,536 masks, corrects the v0.3 effective-placement defect and removes the measured terminal grid. It remains installed under an isolated name; v0.3 is preserved.

Part 0 SHA-256	94847138178994d016d3a0e315be0aa10604c1ea85e01b6d98eb8f41421ac9d8
Part 1 SHA-256	a979a9568dbe5c0b90bb54eb6bc0e4a70caf536bd4c2e791d589796a984f0ec1

Scope boundary

This report treats each stage as a separate gate. A stored outline PASS never claims a terminal PASS. A failed or excluded launch is retained rather than silently discarded.

1 · Fixed mathematical oracle

Virtual x increases left-to-right, but bit significance inside every row increases right-to-left. The number in each square is the bit index.

3	2	1	0
7	6	5	4
11	10	9	8
15	14	13	12

```
terminal_cell_x = vx // 4
terminal_cell_y = vy // 4
local_x = vx % 4; local_y = vy % 4
bit = 4 × local_y + (3 - local_x)
bit_value = 1 << bit
```

Part 0: mask 0000–7FFF → U+F0000 + mask

Part 1: mask 8000–FFFF → U+100000 + (mask – 0x8000)

Machine oracle: 65,536 masks, 65,536 unique codepoints, zero round-trip failures.

2 · Evidence gates

Each gate states what it proves and, equally importantly, what it does not prove.

Gate	Requirement	Observed result
A	Mathematics	PASS · approved separately
B	Stored cmap/outlines/hmtx	PASS · 65,536/65,536
C	Integer-grid FreeType	PASS · 0 mismatches
D	Linux selection and shaping	PASS · no unknown glyphs
E	8–20 pt / 96 DPI solid field	C1/C2 FAIL · C3 PASS
F	14 pt real MATE Terminal	C1 5,632 low pixels · C3 0
G	Independent Linux rebuild	PASS · byte-identical

Why the old conclusion changed

Expectation. A raw outline at x=375 should render at x=375.

Observation. The earlier verifier inspected raw components but did not apply the TrueType hmtx placement formula.

Outcome. The gap in the verifier is now a permanent regression test.

3 · Reproduced v0.3 root cause

TrueType positions a simple glyph with:

```
effective_x = raw_x - (xMin - leftSideBearing)
effective_xMin = leftSideBearing
```

mask	required x	raw xMin	v0.3 LSB	effective xMin
0008	0	0	0	0
0004	1	125	0	0
0002	2	250	0	0
0001	3	375	0	0

Exactly 4,095 nonempty Part 0 masks have xMin > 0 and therefore shift left in v0.3. Part 1 always includes bit 15 in the leftmost column, so its xMin is zero.

Root-cause gate

Expectation. Effective placement equals approved local x.

Observation. v0.3 fails 4,095 masks; candidate.1 fails zero.

Outcome. ROOT CAUSE CONFIRMED.

4 · Candidate.1: corrected exact core

Candidate.1 removes composites, writes the selected-cell union as simple contours and assigns each nonempty glyph an LSB equal to its xMin.

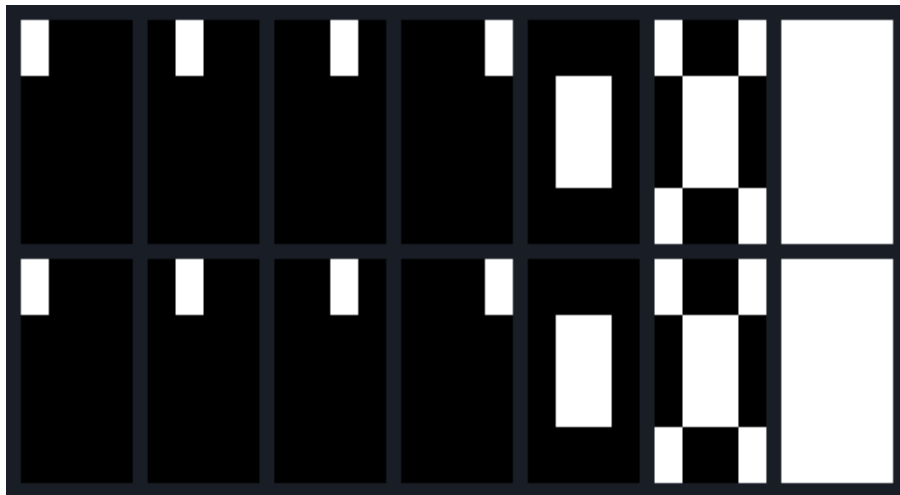
```
500 × 1000 unit cell  
125 × 250 unit subcells  
0 units overfill  
simple union contours  
fixed 500-unit advance
```

Stored-font proof

Expectation. Every cmap, contour edge, decoded mask and metric matches an independent oracle.

Observation. 65,536 patterns; zero outline, mask, metric or effective-position mismatches.

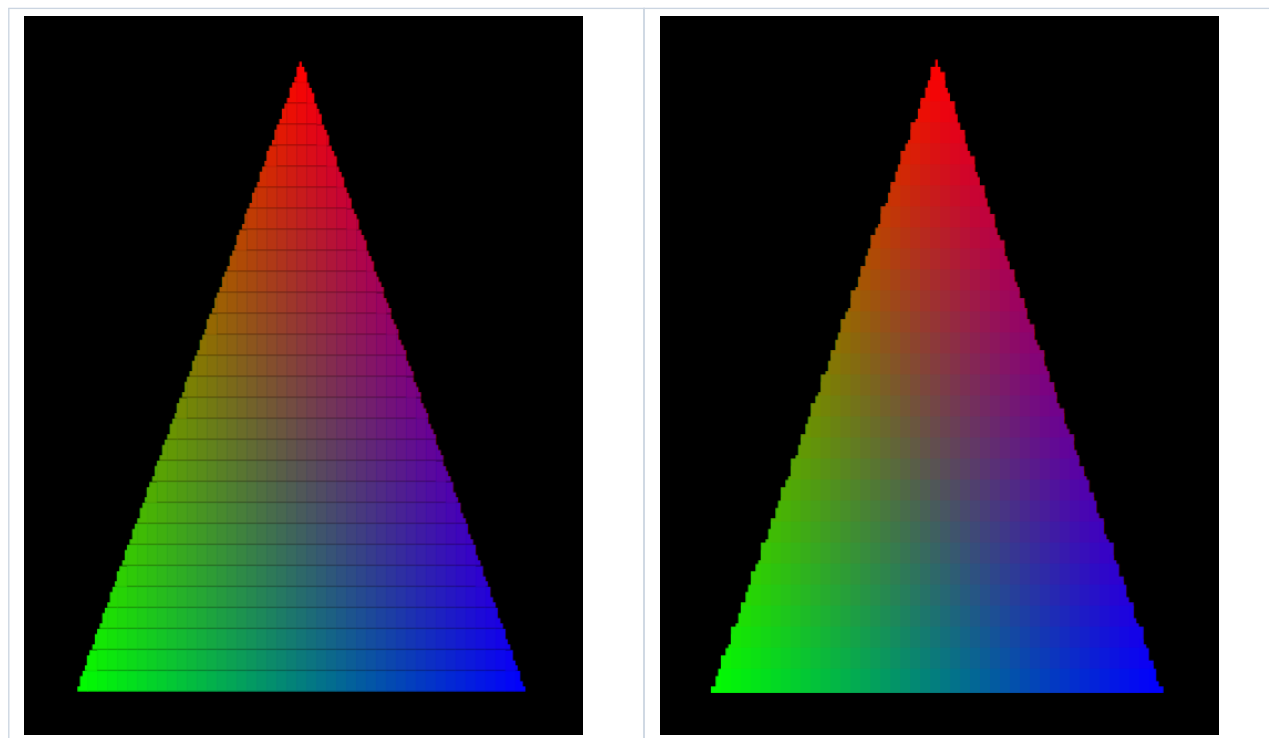
Outcome. PASS · stored data only.



Integer-grid raster: 62 cases at 40, 80 ppem; zero mismatched or partially covered pixels.

5 · The real fractional-raster failure

The integer-grid result did not predict a 14-point, 96-DPI terminal. The nominal 18.666... pixel em is placed in a 19-pixel terminal row, leaving partially covered cached glyph edges.



Candidate.1 · exact vector core

Candidate.3 · 100-unit seam guard

Candidate.1 boundary pixels	5,632 below 90% coverage (16.923%)
Candidate.3 boundary pixels	0 below 90% coverage

The low-coverage pixels concentrate at character boundaries, proving a raster-edge problem rather than a bit-mask problem.

6 · Candidate.2: controlled hinting test

Candidate.2 differs from candidate.1 only in names, native point-grid instruction programs, integer-ppem metadata, the gasp table and maxp instruction maxima. All 65,536 cmap entries, raw contours and hmtx records compare equal.

```
Expected: native grid execution removes fractional edge coverage.  
Observed: 8pt=8,800 · 9pt=45,280 · 10pt=80,000 · 11pt=22,920 · 12pt=61,440 · 13pt=105,600 ·  
14pt=29,480 · 16pt=85,680 · 18pt=93,760 · 20pt=22,880
```

Candidate.2 decision

Expectation. Zero nonwhite seam pixels at each point size.

Observation. Every measured count is identical to candidate.1.

Outcome. FAIL · retained as a negative result; not selected.

This test is tied to the final candidate.2 SHA-256 file and a separate Pango JSON. It is not inferred from a screenshot.

7 · Measured seam-guard threshold

Each probe changed only U+107FFF and its matching bearing. '0' means zero nonwhite pixels; 'x' means a partially covered seam remained.

guard units	8	9	10	11	12	13	14	16	18	20	all
0	x	x	x	x	x	x	x	x	x	x	—
8	x	x	x	x	x	x	x	x	x	x	—
16	x	x	x	x	x	x	x	x	x	x	—
24	x	x	x	x	x	x	x	x	x	x	—
32	x	x	x	x	x	x	x	x	x	0	—
40	x	x	x	x	x	x	0	x	0	0	—
48	x	x	x	0	x	x	0	0	0	0	—
64	x	x	x	0	0	0	0	0	0	0	—
80	x	0	0	0	0	0	0	0	0	0	—
100	0	0	0	0	0	0	0	0	0	0	PASS

100 font units is the first tested value that passes all ten point sizes. At 14 pt, 40 units already passes; the selected value covers the full 8–20 pt test range.

8 · Candidate.3: selected transform

Candidate.3 retains the exact 4×4 core and expands only outline coordinates already lying on a terminal-cell exterior edge.

candidate.1 coordinate	candidate.3 coordinate
x = 0	x = −100
x = 500	x = 600
y = −200	y = −300
y = 800	y = 900
all other x/y	unchanged

Exhaustive derivation proof

Expectation. Every candidate.3 coordinate equals the declared transform of candidate.1; cmap and formula never change.

Observation. 65,536 codepoints compared; zero unexpected outline changes; zero formula/cmap changes.

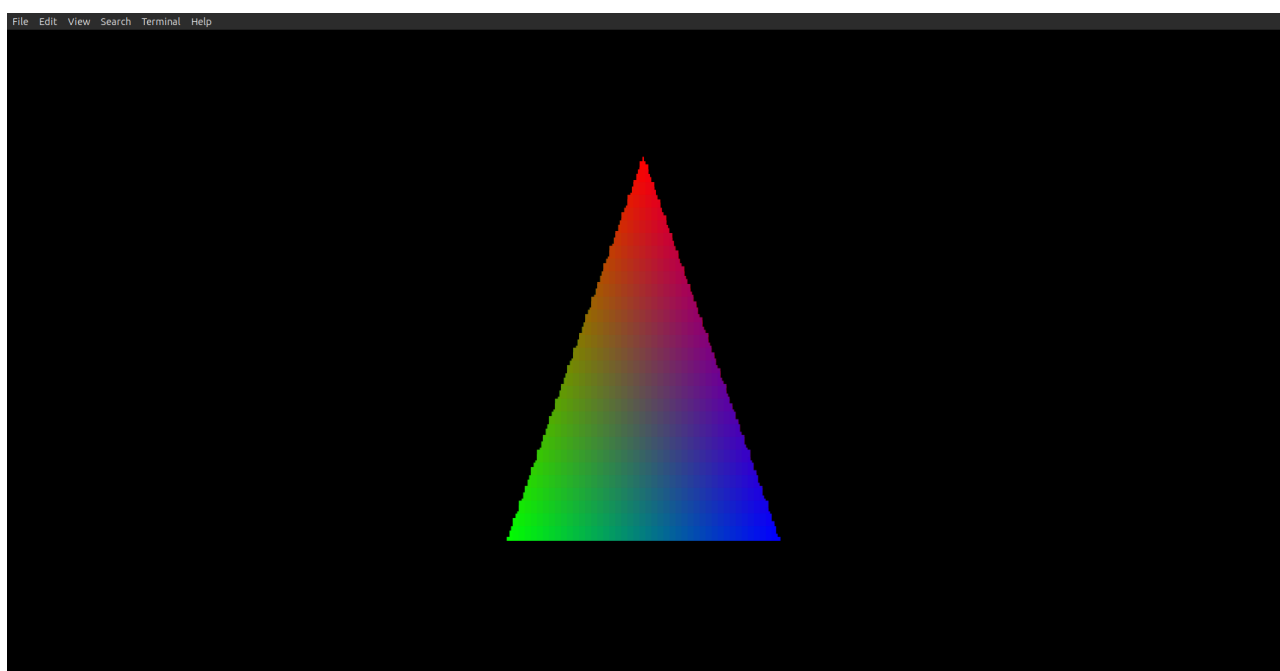
Outcome. PASS.

Guarded glyphs: Part 0 = 32,752; Part 1 = 32,768.

9 · Candidate.3 raster and terminal result

The isolated point-unit renderer and the real terminal now agree.

size	black	nonwhite	status
8	0	0	PASS
9	0	0	PASS
10	0	0	PASS
11	0	0	PASS
12	0	0	PASS
13	0	0	PASS
14	0	0	PASS
16	0	0	PASS
18	0	0	PASS
20	0	0	PASS



Actual 1920×990 MATE Terminal capture · 211×50 cells · 14 pt · 96 DPI

Measured full-mask boundary pixels below 90% expected coverage: 0.

10 · Declared cost: exterior one-bit overhang

A seamless guard is not mathematically free. The exact difference is measured with one tested glyph in the middle of three cells at a 40-pixel em.

mask	candidate.1 run	candidate.3 run	difference
0x0008	[19, 26]	[15, 26]	left 4, right 0
0x0004	[24, 31]	[24, 31]	left 0, right 0
0x0002	[29, 36]	[29, 36]	left 0, right 0
0x0001	[34, 41]	[34, 44]	left 0, right 3

The two interior x positions are unchanged. The exterior positions overhang into the neighboring terminal cell. This is the deliberate implementation difference from the ideal 4×4 core and must be considered by applications drawing isolated boundary pixels.

Trade-off

Expectation. Exact isolated subcell bounds and universal fractional-size seams would both pass.

Observation. The tested raster stack cannot satisfy both with unhinted cached glyphs. Embedded hinting did not change the result; 100-unit overlap does.

Outcome. Candidate.3 prioritizes the user's seamless solid-mass requirement and records the overhang exactly.

11 · Reproducibility and preserved assets

Candidate.3 was regenerated independently on Linux with Python 3.12.3 and FontTools 4.46.0. Both Linux files compare byte-for-byte with the macOS-generated files.

```
Part 0 94847138178994d016d3a0e315be0aa10604c1ea85e01b6d98eb8f41421ac9d8
Part 1 a979a9568dbe5c0b90bb54eb6bc0e4a70caf536bd4c2e791d589796a984f0ec1
```

The repository package was then installed on Linux from its checked-in directory rather than from a build tree. Installed bytes matched the package; wwidth was one throughout both ranges; Fontconfig selected both candidate parts; and Pango shaped text plus P0/P1 with zero unknown glyphs.

The unversioned launcher and every named demo now select the candidate-specific profile. Linux readback proved `PUA 4x4 v0.4 Candidate 3 12`; installed P0/P1 hashes and Fontconfig selections matched RC1. Demo masks, Boolean operations and codepoint formulas were not changed. `PUA4X4\_USE\_V03=1` retains an explicit legacy route.

v0.3, candidate.1, candidate.2 and candidate.3 all have distinct files/families. The current `PUA 4x4` alias was not overwritten.

Recorded procedural exception

The first candidate.3 launch inherited `visible-name='Default'`; MATE reported `No such profile "(null)", using default profile`. That window was excluded. The name was corrected and read back with font `PUA 4x4 v0.4 Candidate 3 14` before the accepted capture. Both logs are retained.

12 · Reproduce and inspect

```
cd experiments/pua-4x4
python3 verify_mathematical_mapping.py
python3 generate_pua4x4_v04_candidate.py
python3 verify_pua4x4_v04_candidate.py
python3 audit_horizontal_placement.py
python3 verify_candidate_freetype_raster.py
python3 make_v04_candidate2_hinted.py
python3 verify_v04_candidate2_derivation.py
python3 make_fullmask_overfill_probes.py
python3 make_v04_candidate3_seamguard.py
python3 verify_v04_candidate3_derivation.py
python3 verify_v04_rc1_package.py
./install-linux-v04-candidate3.sh
./launch-linux-v04-candidate3.sh triangle
```

Primary machine-readable evidence

```
pua4x4-horizontal-placement-root-cause.json
pua4x4-v0.4-candidate-font-audit.json
pua4x4-v0.4-candidate-freetype-raster.json
pua4x4-v0.4-candidate2-final-diagnostic-pango-raster.json
fullmask-overfill-probes/measurements.json
pua4x4-v0.4-candidate.3-derivation.json
pua4x4-v0.4-candidate3-pango-raster.json
pua4x4-v0.4-terminal-triangle-comparison.json
pua4x4-v0.4-candidate3-onebit-overhang.json
pua4x4-v0.4-rc1-packaged-linux-runtime.json
pua4x4-v0.4-demo-launcher-profile.json
cross-environment/candidate3-linux-hashes.txt
```

Decision state

Candidate.3 is the recommended v0.4 release candidate for visual acceptance. Promotion to the unversioned `PUA 4x4` alias is intentionally separate from this proof so the preserved v0.3 environment cannot be mistaken for the tested candidate.